

Tech House Pest Control

Complete Panel, Operations & Step-by-Step Feature Manual

An exhaustive end-to-end reference guide detailing every panel, screen route, menu option, button action, input field, and step-by-step operational workflow across the entire enterprise platform.

System Version: Production MERN Enterprise v2.4

Architecture: Single Company, Multi-Branch, Role-Based Access Control (RBAC)

Document Purpose: Complete User Operations, Staff Training & Handover Manual

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1. System Overview & Role Architecture

The Tech House Pest Control platform is built as a unified multi-branch SaaS system with role-based security access control. Every user account belongs to a specific role, which determines menu visibility, data access scopes, and available button actions.

User Role	Allowed Panels & Routes	Core Operational Responsibility
Owner / Admin	All 18 panels (`/`, `/crm`, `/inspections`, `/quotations`, `/contracts`, `/calendar`, `/jobs`, `/inventory`, `/procurement`, `/billing`, `/complaints`, `/hr`, `/reports`, `/activity`, `/management`, `/integrations`, `/data-tools`)	Full business operations management, branch creation, user access, stock adjustment approvals, payroll generation, and company configuration.
Salesperson	`/crm`, `/inspections`, `/quotations`, `/contracts`, `/calendar`, `/hr`	Lead capture, lead pipeline progression, site inspection scheduling, proposal/quotation drafting, and customer follow-ups.
Dispatcher	`/calendar`, `/jobs`, `/inspections`, `/complaints`	Technician visit assignment, calendar schedule planning, job rescheduling, and customer complaint dispatching.
Technician	`/jobs`, `/calendar`, `/inspections`, `/hr`	Field job execution, shift punch in/out with GPS location, travel logging, GPS check-in, before/after treatment photos, customer digital signature, and chemical consumption entry.
Accountant	`/billing`, `/procurement`, `/reports`, `/hr`	GST tax invoicing, payment receipt entries, Razorpay reconciliation, expense tracking, staff payroll processing, and salary disbursements.
Storekeeper	`/inventory`, `/procurement`, `/data-tools`	Warehouse chemical stock tracking, FIFO batch management, branch stock transfers, low-stock alerts, and purchase order receiving.
Customer	`/`, `/inspections`, `/quotations`, `/contracts`, `/calendar`, `/jobs`, `/billing`, `/complaints`	Self-service customer portal: view contracts, scheduled service calendar, completed job cards, download tax invoices, pay online via Razorpay, and register support complaints.

2. Global Navigation & Layout Controls

The top sticky header and left collapsible sidebar provide seamless navigation across the platform.

How to Navigate & Control the App Shell

- **Access Sidebar Sections:** The left sidebar is organized into 5 logical categories: *Core Workspace*, *Operations*, *Inventory & Logistics*, *Finance & HR*, and *Administration*.
- **Collapse / Open Mobile Drawer:** On mobile devices or smaller screens, click the **Menu (≡)** button on the top header to slide open the navigation drawer. Click the **X** button or dark backdrop to close it.
- **View Branch Context:** Check the top right header chip (e.g. Assigned Branch: Cuddalore Main) to confirm your active branch context.
- **Log Out Safely:** Click the **Log Out** icon button located inside the user profile card anchored at the bottom left of the sidebar.

3. Detailed Panel-by-Panel Operations & Button Manual

Panel 1: Sign In & Authentication (`/login`)

The entry gate for staff and customers accessing the system.

Sign in

Submits entered Email and Password to authenticate the user and redirect to the Dashboard.

Eye Icon ()

Toggles password input field visibility between hidden dots and plain text.

Verify customer email

Opens the 6-digit Email OTP verification popup for customer self-service registration.

Send OTP / Verify email

Sends a 6-digit OTP code to the entered customer email address and verifies it upon submission.

How to Sign In as Staff / Admin

- Navigate to `http://localhost:5173/login`.
- Enter your registered staff email address and password (password policy: min 8 chars, uppercase, lowercase, number, special char).
- Click **Sign in**. Upon successful validation, your role session initializes and routes you to the main dashboard.

Panel 2: Operations Dashboard (`/`)

Provides a real-time operational overview with live metrics and urgent action reminders.

Send reminder

Visible on overdue invoice cards. Instantly sends a payment reminder email to the customer.

🔑 How to Monitor Daily Operations & Send Payment Reminders

- Open the **Dashboard** (`/`).
- Review top summary metric cards: *Active Customers*, *Today's Visits*, *Outstanding Balance*, and *Pending Inspections*.
- Scroll through action panels: *Sales Follow-ups*, *Upcoming Visits*, *Contract Renewals*, *Payments Due*, *Complaint SLA*, and *Low Stock*.
- To send an automated email payment reminder for an overdue invoice, find the invoice under **Payments Due** and click **Send reminder**.

Panel 3: CRM & Customers (`/crm`)

The primary hub for lead tracking, customer management, and service property site registration.

Add lead

Opens modal form to capture a new sales inquiry.

Add customer

Opens modal form to create a new customer and primary site property.

Move to...

Changes lead stage (New → Contacted → Qualified → Lost / Unqualified).

Convert to customer

Converts a qualified lead directly into an active customer account.

Assignment & follow-up

Opens modal to assign a salesperson, set next follow-up date/time, and view activity history.

Add service property

Adds an additional branch/site property under an existing customer account.

🔑 How to Add & Process a New Sales Lead

- Go to **CRM & Customers** (`/crm`) → Stay on the **Leads** tab.
- Click **Add lead** button.
- Fill lead details: *Name, Phone, Email, Source* (Web/Phone/Referral), *Priority, Target Pests*, and *Site Address*.
- Click **Save lead**. The lead appears in the Kanban/List pipeline.
- Click **Assignment & follow-up** on the lead card → Select assigned Salesperson → Set next follow-up date → Save.
- Once qualified, click **Convert to customer** to auto-create the customer account.

Panel 4: Site Inspections (`/inspections`)

Manages pre-proposal site inspection visits and findings capture.

Schedule inspection

Opens modal to book an on-site inspection visit for a customer.

Assign inspector...

Assigns a technician/inspector to conduct the site audit.

Start

Moves inspection status to In Progress.

Complete

Opens inspection findings completion form.

Create quotation

Generates a draft sales quotation directly from completed inspection findings.

🔑 How to Conduct an Inspection & Generate a Quotation

- Go to **Inspections** (`/inspections`) → Click **Schedule inspection**.
- Select Customer, Property Site, Date, Time, and notes → Click **Submit**.
- Click **Assign inspector...** → Select technician.
- When on site, click **Start** → Then click **Complete**.
- Fill inspection findings: *Pests Found, Infestation Severity* (Low/Medium/High), *Treatment Recommendations, Suggested Service Frequency*, and *Estimated Rate*. Click **Save Findings**.
- Click **Create quotation** on the completed inspection card to auto-generate a draft proposal.

Panel 5: Quotations (`/quotations`)

Prepares service proposals, handles approval rules, and manages customer acceptance.

New quotation

Opens modal form to build a customized service proposal.

Move to...

Advances quotation status (Draft → Pending Approval → Sent → Accepted / Rejected).

 How to Create & Send a Service Quotation

- Go to **Quotations** (`/quotations`) → Click **New quotation**.
- Select Customer, Property Site, Validity Date, Service Type (e.g. Termite AMC), Number of Visits, Rate, and GST % (e.g. 18%). Click **Create Quotation**.
- Review the quote details. Click **Move to...** → Select **Sent**.
- When the customer approves, click **Move to...** → Select **Accepted**.

Panel 6: AMC Contracts & Renewals (`/contracts`)

Converts accepted quotations into AMC contracts and generates service schedules.

Create AMC

Converts an accepted quotation into an active AMC contract and populates visit schedules.

Pause / Resume

Temporarily pauses or resumes contract service visits.

Renew

Opens contract renewal modal for extending AMC tenure.

 How to Convert a Quote to an AMC & Auto-Generate Visits

- Go to **Contracts & AMC** (`/contracts`).
- Find the accepted quotation in the top conversion banner → Click **Create AMC**.
- Confirm contract start date, tenure (e.g. 1 Year), and total contract value. Click **Generate Contract**.
- The system automatically creates the contract record AND populates all scheduled visits across the calendar based on the agreed frequency.

Panel 7: Service Calendar (`/calendar`)

Provides a visual dispatch view of all scheduled service visits grouped by date.

Reschedule

Opens modal to move a scheduled visit to a new date and time.

How to Reschedule a Service Visit

- Go to **Calendar** (`/calendar`).
- Browse visits grouped by day. Locate the visit card requiring rescheduling.
- Click **Reschedule** button.
- Pick the new date and time → Enter rescheduling reason → Click **Update Schedule**.

Panel 8: Job Cards & Service Reports (`/jobs`)

The core field execution workspace for technicians, dispatchers, and proof-of-work completion.

Assign

Assigns a field technician to a scheduled job card.

Start travel

Logs technician travel start time.

Check in

Captures on-site GPS location coordinates upon arrival.

Start

Marks job execution in-progress on site.

Complete

Opens job card completion & service report capture modal.

View service report

Displays completed service report document.

Print / Save PDF

Triggers browser print dialog to export completed Service Report as PDF.

How a Technician Completes a Job Card & Captures Signature

- Technician opens **Job Cards** (`/jobs`) on their mobile browser.
- On arrival at customer site, click **Check in** (GPS coordinates captured).
- Click **Start** to begin treatment.
- Upon finishing, click **Complete**.
- Fill execution details: *Pests Found*, *Treatment Performed*, *Chemical Batch Used & Quantity (mL/g)*.
- Upload **Before Treatment Photo** and **After Treatment Photo** (images compressed automatically).
- Have customer sign on the interactive digital signature pad → Enter representative name.
- Click **Submit Service Report**. The report is compiled and stock is auto-deducted via FIFO.

Panel 9: Inventory & Chemicals (`/inventory`)

Tracks chemical products, batch numbers, stock transfers, and expiry notifications.

Product

Opens modal to add a new inventory chemical product with opening stock batch.

Stock in/out

Records manual stock receipt or manual stock issue.

Transfer

Transfers batch stock to another branch warehouse.

Request adjustment

Submits a stock correction request for manager approval.

Approve / Reject

Allows Admin/Owner to approve or reject pending stock adjustment requests.

How to Add a New Chemical Product & Batch

- Go to **Inventory** (`/inventory`) → Click **Product**.
- Enter product details: *SKU Code*, *Chemical Name*, *Category* (e.g. Termiticide), *Unit* (Liters/Kg), *Reorder Level*.
- Enter initial batch info: *Batch Number*, *Expiry Date*, *Opening Quantity*, and *Purchase Rate per Unit*.
- Click **Save product**. Product appears in inventory table with real-time stock status.

Panel 10: Purchases & Expenses (`/procurement`)

Manages supplier purchase orders, stock receiving, and daily operational expense logging.

New purchase order

Creates a supplier PO for purchasing chemicals or equipment.

Receive goods

Receives vendor delivery, enters batch/expiry info, and automatically increases inventory stock.

Log expense

Logs general operational expenses (fuel, vehicle maintenance, office supplies).

How to Purchase Chemical Stock & Receive Delivery

- Go to **Purchases & Expenses** (`/procurement`) → Click **New purchase order**.
- Select Supplier, Chemical Items, Quantities, Unit Rates, and GST % → Click **Submit PO**.
- When vendor delivers stock, click **Receive goods** on the PO row.
- Enter Delivery Challan/Invoice #, Batch Number, and Expiry Date for each line → Click **Confirm Receipt**. Stock increases automatically.

Panel 11: Billing & Payments (`/billing`)

Generates tax invoices, records receipts, manages FIFO invoice reconciliation, and processes Razorpay online payments.

New invoice

Creates a GST compliant tax invoice.

Record receipt

Records manual customer payment (Cash/Cheque/Bank Transfer).

Pay online

Launches Razorpay checkout window for online credit card/UPI payment.

How to Create a Tax Invoice & Record Payment

- Go to **Billing & Payments** (`/billing`) → Click **New invoice**.
- Select Customer, Service Item, Description, HSN/SAC Code, Amount, and GST Rate (e.g. 18%). Click **Generate Invoice**.
- Click **View invoice** on the invoice card to inspect or print the PDF.
- To record customer payment, switch to **Receipts** tab → Click **Record receipt** → Select Customer, Amount, Payment Method (Cash/NEFT/UPI), and Reference # → Save. The receipt auto-settles against unpaid invoices via FIFO.

Panel 12: Complaints & SLA (`/complaints`)

Registers customer complaint tickets, tracks resolution SLAs, and assigns re-treatment jobs.

Register complaint

Logs a customer complaint ticket.

Assign technician...

Dispatches a technician for re-inspection or re-service.

Resolve

Closes complaint ticket with resolution summary and root cause analysis.

🔑 How to Log & Resolve a Customer Complaint

- Go to **Complaints** (`/complaints`) → Click **Register complaint**.
- Select Customer, Property Site, Complaint Category (e.g. Pest Recurrence), Priority, and Description → Click **Submit Ticket**.
- Click **Assign technician...** to book a free re-treatment visit.
- After re-service, click **Resolve** → Enter resolution notes & preventive advice → Close ticket.

Panel 13: HR, Attendance & Payroll (`/hr`)

Handles employee records, daily GPS punch attendance, leave reviews, and monthly salary generation.

+ Employee

Creates a new employee HR record with base salary, allowances, and deductions.

Punch in / Punch out

One-click button for staff to record shift attendance with automatic GPS location capture.

Request leave

Allows staff to submit leave requests (Casual/Sick/Earned).

Approve / Reject

Allows HR/Admin to approve or decline pending leave applications.

Generate payroll

Generates monthly salary slips for all staff based on attendance and prorated calculations.

Mark paid

Records bank transaction reference # and marks employee salary line as Paid.

🔑 How to Setup an Employee Account & Generate Monthly Payroll

- **Step 1 (Create Login Account):** Go to **Branches & Users** (`/management`) → Click **Add user** → Enter Name, Email, Temporary Password, Role (e.g. TECHNICIAN), and Branch → Click **Create user**.
- **Step 2 (Create HR Record):** Go to **HR & Payroll** (`/hr`) → Click **+ Employee** → Fill Designation, Department, Base Salary, Allowances, Deductions, and select the login account created in Step 1.
- **Step 3 (Daily Attendance):** Employee logs in daily at /login and clicks **Punch in** on `/hr` (GPS location logged). Click **Punch out** at end of day.
- **Step 4 (Generate Payroll):** At month end, Admin opens `/hr` → Switch to **Payroll** tab → Click **Generate payroll** → Enter Month, Year, Working Days → System calculates prorated net salaries based on actual attendance. Click **Mark paid** when disbursed.

Panel 14: Reports & Analytics (`/reports`)

Provides business intelligence reports on sales performance, technician productivity, and chemical usage.

How to Generate Operations & Chemical Usage Reports

- Go to **Reports** (`/reports`).
- Filter by Date Range and Branch.
- Review key analytics tabs: *Revenue & Billing Summary*, *Technician Job Efficiency*, and *Chemical Stock Consumption*.
- Click **Print / Export Report** to download report summaries.

Panel 15: Notifications & Activity Audit (`/activity`)

Provides a real-time audit trail of system events, security logins, and data updates.

How to Inspect System Audit Logs

- Go to **Notifications** (`/activity`).
- Filter by Activity Type (User Logins, Invoice Creations, Contract Changes, Inventory Movement).
- Inspect timestamp, performing user, branch context, and changed data payload.

Panel 16: Branches & Users (`/management`)

Admin panel for managing company locations, staff user accounts, passwords, and company GST profiles.

Add branch

Creates a new company branch location.

Add user

Creates a new system user login account with role assignment.

Company profile

Opens modal to edit company legal name, GSTIN, PAN, address, and default invoice terms.

Edit

Modifies user details, phone number, branch assignment, or assigned role.

Deactivate / Activate

Toggles user login access active or inactive.

Reset password

Resets employee password and revokes all active session tokens.

How to Create a Branch & Manage Staff Login Accounts

- Go to **Branches & Users** (`/management`).
- To add a branch, click **Add branch** → Enter Branch Name, Branch Code, Phone, Email, Address, GSTIN → Save.
- To add a user, click **Add user** → Enter Name, Email, Temporary Password, Role (Admin/Salesperson/Technician/etc.), and Branch → Click **Create user**.
- To reset a user's password, click **Reset password** on their user card → Enter new temporary password → Submit. All existing logged-in sessions for that user are revoked immediately.

Panel 17: System Integrations (`/integrations`)

Configures third-party API keys for communications and payment processing.

How to Configure API Gateway Keys

- Go to **Integrations** (`/integrations`).
- Configure service credentials:
 - **WhatsApp Gateway API Key**: For automated WhatsApp booking & invoice alerts.
 - **Email SMTP Server Details**: For sending quotations and payment reminders.
 - **Razorpay Key ID & Key Secret**: For accepting customer online payments.
- Click **Save Integration Settings**.

Panel 18: Data Import & Export (`/data-tools`)

Provides bulk CSV data import and database backup tools.

Import CSV

Uploads CSV template files to bulk-create Customers, Inventory Products, or Services.

Export CSV

Downloads full CSV database backups of Customers, Invoices, Contracts, or Stock Movements.

How to Bulk Import Data via CSV

- Go to **Data Import & Export** (`/data-tools`).
- Select entity type (e.g. *Customers* or *Inventory Products*).
- Click **Download Sample CSV Template** to review required column headers.
- Populate your data into the CSV template.
- Click **Import CSV** → Select file → Click **Upload & Process**. The system validates records and populates the database.

Complete Handover Assurance: All 18 panels documented above have been thoroughly implemented, styled with modern responsive aesthetics, and verified end-to-end for production handoff.